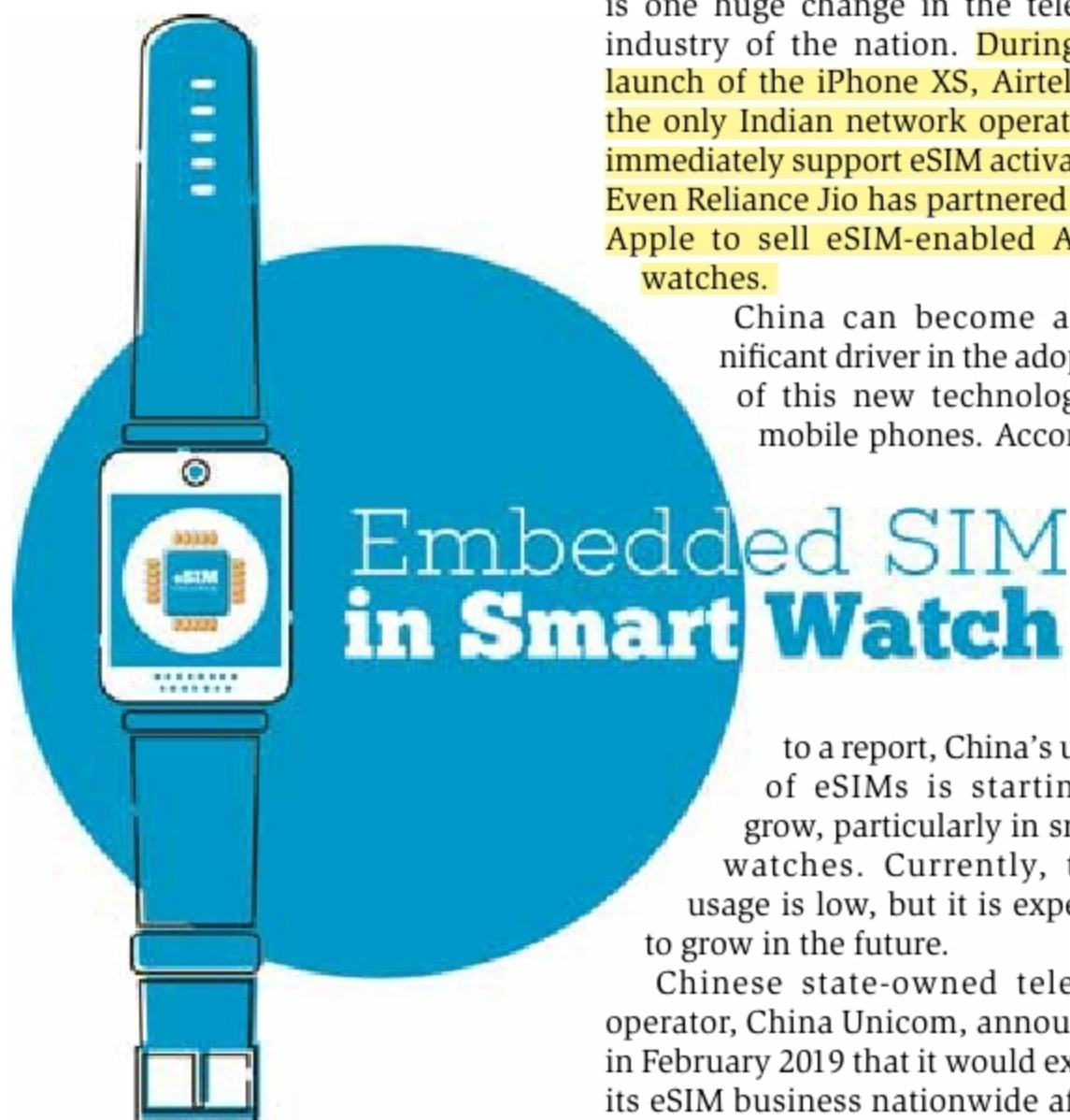


We may get the first device without traditional SIM cards in 2020. eSIM needs to be supported by a carrier or network. Though XR, XS, and XSMax have physical SIM slots, eSIM inside these handsets is used as second SIM support. eSIM is backed by GSMA. iPad Pro, Apple watch series 3, and iPhone XI are planning to move completely to eSIMs, and this could happen in 2020.

Google's Pixel 2, Pixel 3, and Pixel 3XL also support eSIMs. eSIMs are



also checked on computers and laptops to reveal the wireless connection.

Some Windows 10 devices also support eSIM that lets you connect to a cellular data network. eSIMs are likely to play a key role in the next generation home-based LTE connected computers. You can have more than one eSIM on your phone, but you can use only one of these at a time.

With eSIM supported smartphones, you need a carrier app or a QR code that you can scan. In the UK, EE (Everything Everywhere) and Vodafone support eSIM, although Vodafone is currently only supporting it for Apple Watch Series 3, 4, or

5 Cellular. In the US, AT&T, T-Mobile, and Verizon Wireless support eSIM.

Apart from traditional SIM service providers, the global mobile network Truphone has also started cellular eSIM data plans. These plans can be operated using the MyTruphone app. Truphone's international plans work across eighty countries, including in Europe, America, and Australia.

Airtel is one of the first telecom operators from India to introduce eSIM technology in the country. This is one huge change in the telecom industry of the nation. During the launch of the iPhone XS, Airtel was the only Indian network operator to immediately support eSIM activation. Even Reliance Jio has partnered with Apple to sell eSIM-enabled Apple watches.

China can become a significant driver in the adoption of this new technology in mobile phones. According

to a report, China's usage of eSIMs is starting to grow, particularly in smartwatches. Currently, their usage is low, but it is expected to grow in the future.

Chinese state-owned telecom operator, China Unicom, announced in February 2019 that it would extend its eSIM business nationwide after a long trial in seven cities. The telecom giant has said that it has built an eSIM management system using its own, self-developed technology. The company also announced that it has integrated its eSIM technology with more than twenty intelligent devices, including Apple Watch 4. All three Chinese SIM hardware manufacturers—China Mobile, China Telecom, and China Unicom—have launched their respective eSIM solutions.

Transforming subscribers' identities and Internet management

The humble SIM card has survived for quite a long period but finally looks

like the era of SIM cards is on its way out. Why?

It's just because of eSIM, a much smarter technology. This new technology is surely going to be the future of SIM technology.

For IoT platforms, eSIMs can bring a greater change. The Hyundai Venue model car has already got an eSIM feature. Growing awareness about wearable devices and surge in mobile computing are pushing up the eSIM market. However, the lack of regulations and concerns related to security may impede eSIM market growth for next 3-4 years.

"eSIM market in the Asia Pacific may grow faster over the next five years due to the presence of favourable government regulations for telecommunications," says Karan Chechi, research director at TechSci Research.

Applications

- Smartphones and smartwatches
- Wireless security cameras
- LTE operated computers and laptops
- Smart meters used for billing, power management, and water management system, etc
- Home appliances such as refrigerators, air-conditioners, etc
- Several other IoT connected devices such as fingerprint scanners
- In M2M communication

To sum up

The eSIM market is still in its nascent stage. eSIM penetration in India is less than one per cent and is expected to grow by 25 per cent by 2025. Globally, the eSIM market is estimated to grow to 978.3 million dollars in 2023 from 253.8 million dollars in 2018.

An in-depth study and analysis are required for further developments in eSIM. Recent technologies like artificial intelligence (AI), machine learning (ML), and Big Data can be used to make further improvements in eSIM.

Several giants like AT&T, China Telecom, Airtel, Vodafone, and other SIM service providing companies are trying to bring several innovations in this sector. Thus, next-generation smartphone development has already begun with the slow but steady emergence of eSIMs. **EFY**